

A Proof of the Second Fundamental Theorem of Calculus

Statement of the Theorem

Theorem (Second Fundamental Theorem of Calculus). *Let f be a continuous real-valued function on the closed interval $[a, b]$, and let F be any antiderivative of f on $[a, b]$, meaning that F is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $F'(x) = f(x)$ for all $x \in (a, b)$. Then*

$$\int_a^b f(x) dx = F(b) - F(a).$$

Remark. This is sometimes called the *Evaluation Theorem*. It is the form of the Fundamental Theorem of Calculus that allows one to compute a definite integral by evaluating an antiderivative at the endpoints.

Tools Used in the Proof

The proof relies on three key results from analysis:

1. **Mean Value Theorem (MVT).** If g is continuous on $[\alpha, \beta]$ and differentiable on (α, β) , then there exists $c \in (\alpha, \beta)$ such that

$$g(\beta) - g(\alpha) = g'(c) (\beta - \alpha).$$

2. **Riemann integrability of continuous functions.** Every function continuous on a closed interval $[a, b]$ is Riemann integrable on $[a, b]$.
3. **Definition of the Riemann integral.** If f is Riemann integrable on $[a, b]$, then for every sequence of tagged partitions $\{(P_n, \{c_i^{(n)}\})\}$ with mesh $\|P_n\| \rightarrow 0$,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{k_n} f(c_i^{(n)}) \Delta x_i^{(n)} = \int_a^b f(x) dx.$$

Proof

Proof. Let $\varepsilon > 0$ be arbitrary, and let

$$P = \{x_0, x_1, x_2, \dots, x_n\}, \quad a = x_0 < x_1 < \dots < x_n = b,$$

be any partition of $[a, b]$. Write $\Delta x_i = x_i - x_{i-1}$ for the width of the i -th subinterval, and $\|P\| = \max_{1 \leq i \leq n} \Delta x_i$ for the mesh of P .

Step 1: Telescoping. We begin by writing $F(b) - F(a)$ as a telescoping sum across the partition:

$$F(b) - F(a) = F(x_n) - F(x_0) = \sum_{i=1}^n [F(x_i) - F(x_{i-1})].$$

Step 2: Applying the Mean Value Theorem. By hypothesis, F is continuous on $[a, b]$ and differentiable on (a, b) , so for each $i \in \{1, 2, \dots, n\}$, F is continuous on $[x_{i-1}, x_i]$ and differentiable on (x_{i-1}, x_i) . The Mean Value Theorem therefore guarantees the existence of a point $c_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = F'(c_i)(x_i - x_{i-1}) = f(c_i) \Delta x_i,$$

where in the last equality we used $F'(c_i) = f(c_i)$.

Step 3: Identifying a Riemann sum. Substituting the result of Step 2 into the telescoping sum from Step 1 gives

$$F(b) - F(a) = \sum_{i=1}^n f(c_i) \Delta x_i. \quad (\star)$$

The right-hand side of $(\star)$ is precisely a Riemann sum for f over $[a, b]$, with tags c_i chosen (via the MVT) from the open subintervals (x_{i-1}, x_i) .

Step 4: Passing to the limit. Because f is continuous on the closed interval $[a, b]$, f is Riemann integrable on $[a, b]$. Hence for every sequence of partitions P_n of $[a, b]$ with $\|P_n\| \rightarrow 0$, and for any choice of tags, the associated Riemann sums converge to $\int_a^b f(x) dx$. In particular, choose any such sequence $\{P_n\}$, and for each n let $\{c_i^{(n)}\}$ be the tags produced by the Mean Value Theorem as in Step 2. Equation $(\star)$ gives

$$F(b) - F(a) = \sum_{i=1}^{k_n} f(c_i^{(n)}) \Delta x_i^{(n)} \quad \text{for every } n.$$

Taking $n \rightarrow \infty$ and using Riemann integrability,

$$F(b) - F(a) = \lim_{n \rightarrow \infty} \sum_{i=1}^{k_n} f(c_i^{(n)}) \Delta x_i^{(n)} = \int_a^b f(x) dx.$$

The left-hand side does not depend on n , so this equality is exact (not merely a limit). This completes the proof. $\square$

Discussion

A few remarks are in order.

Why the MVT is the right tool. The clever step is Step 2. We do not know a priori that the antiderivative F is related to f in any integrable way; all we know is the pointwise identity $F'(x) = f(x)$. The Mean Value Theorem converts this pointwise (infinitesimal) information into a statement about *finite differences* $F(x_i) - F(x_{i-1})$, which is exactly what appears on the right-hand side of the telescoping sum.

Where continuity of f is used. Continuity of f is used in Step 4 to guarantee that f is Riemann integrable and that Riemann sums converge to the integral regardless of tag choice. The proof can be extended to the broader case in which f is merely assumed to be Riemann integrable (with F still an antiderivative), though that version requires a more delicate argument controlling the partition directly rather than passing through continuity.

Relationship to the First Fundamental Theorem. The First Fundamental Theorem asserts that if f is continuous on $[a, b]$, then the function $G(x) = \int_a^x f(t) dt$ is an antiderivative of f . One can alternatively deduce the Second Fundamental Theorem from the First by noting that any two antiderivatives of f differ by a constant, so $F(x) = G(x) + C$ for some constant C , whence $F(b) - F(a) = G(b) - G(a) = \int_a^b f - 0 = \int_a^b f$. The proof given above is independent of the First Fundamental Theorem and relies only on the Mean Value Theorem together with Riemann integrability of continuous functions.